OSDFace:
One-Step Diffusion Model for Face Restoration

Group 16

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Course: Deep Learning for Computer Vision

Project Title: OSDFace — One-Step Diffusion Model for Face Restoration

Date: April 2026

1 Abstract

Face restoration aims to recover high-quality (HQ) face images from degraded low-quality (LQ) inputs caused by blur, noise, downsampling, and JPEG compression. While diffusion models have shown remarkable performance in image generation and restoration, their multi-step iterative inference remains computationally prohibitive for real-world deployment. This project presents a comprehensive study of **OSDFace**, a novel **one-step diffusion model** specifically designed for face restoration. OSDFace introduces a **Visual Representation Embedder (VRE)** to capture rich facial priors from low-quality inputs, incorporates a **facial identity loss** for identity consistency, and employs a **GAN-based discriminator** to ensure distribution alignment between restored and ground-truth faces. The model achieves state-of-the-art performance on multiple benchmarks while requiring only a **single diffusion step** ($\approx$0.4 seconds for 512×512 images), making it practical for real-world face restoration applications.

2 Introduction

2.1 Problem Statement

Face images captured in real-world conditions frequently suffer from multiple forms of degradation — motion blur, sensor noise, low resolution caused by downsampling, and compression artifacts from JPEG encoding. These degradations make face images unsuitable for downstream tasks such as face recognition, verification, and visual communication. **Face restoration** is the inverse problem of recovering the original high-quality face from its degraded counterpart, which is inherently **ill-posed** because multiple high-quality images can correspond to the same degraded input.

2.2 Challenges in Face Restoration

The key challenges in face restoration include:

1. **Preserving Identity:** The restored face must faithfully preserve the identity of the person, including subtle features like eye shape, skin tone, and facial structure.
2. **Generating Realistic Details:** Fine details such as hair strands, eyelashes, skin texture, and facial accessories must be recovered naturally without artifacts.
3. **Handling Complex Backgrounds:** Real-world face images often contain complex backgrounds that must be coherently reconstructed alongside the face.
4. **Inference Speed:** Practical applications demand fast inference, ruling out methods requiring hundreds of iterative denoising steps.

2.3 Motivation for OSDFace

Traditional generative approaches for face restoration include:

- **GAN-based methods** (e.g., GFPGAN, CodeFormer) — Effective but prone to mode collapse and training instability.
- **Multi-step diffusion models** (e.g., DiffBIR, DifFace, PGDiff) — Produce high-fidelity results but require 50–1000 denoising steps, making them too slow for real-time use.
- **Existing one-step diffusion models** (e.g., OSEDiff) — Designed for general image restoration and not optimized for the unique demands of face-specific features.

OSDFace addresses all these limitations by being the **first one-step diffusion model specifically designed for face restoration**, combining the powerful generative capabilities of diffusion models with the speed of single-step inference.

2.4 Contributions of OSDFace

The key contributions of the OSDFace framework are:

- **First one-step diffusion model for faces** — Leveraging pretrained Stable Diffusion with only the UNet fine-tuned via LoRA.
- **Visual Representation Embedder (VRE)** — A novel module that captures rich facial priors from LQ inputs using vector-quantized dictionaries.
- **Comprehensive face alignment losses** — Including facial identity loss (via ArcFace), edge-aware perceptual loss (EA-DISTS), and GAN loss for distribution alignment.

3 Background

3.1 Denoising Diffusion Probabilistic Models (DDPM)

Paper: Ho, J., Jain, A., & Abbeel, P. (2020). *"Denoising Diffusion Probabilistic Models."* NeurIPS 2020.

3.1.1 What It Does

DDPM is the foundational generative model that OSDFace builds upon. It generates high-quality images by learning to **reverse a gradual noise-addition process**. Starting from pure Gaussian noise, the model iteratively denoises the signal to produce realistic images.

3.1.2 The Directed Graphical Model

The diffusion process is modeled as a **Markov chain** with two phases:

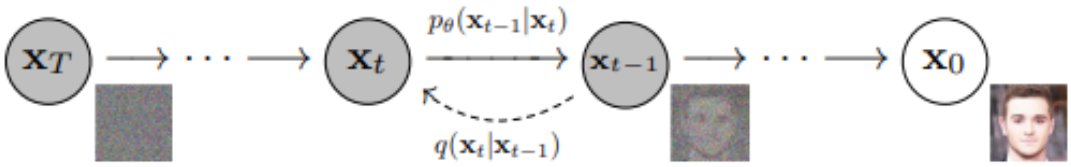


Figure 2: The directed graphical model considered in this work.

Figure 1: DDPM Architecture — The directed graphical model showing the forward diffusion process $q(x_t|x_{t-1})$ progressively adding noise to data x_0, and the reverse process $p_\theta(x_{t-1}|x_t)$ learning to denoise. The forward process transforms clean data into pure Gaussian noise x_T, while the learned reverse process generates data from noise.

3.1.3 How It Works — Forward Process

The **forward (diffusion) process** gradually adds Gaussian noise to data over T timesteps. Given a clean data sample $x_0 \sim q(x_0)$, the forward process produces progressively noisier versions:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1},\ \beta_t \mathbf{I}) \tag{1}$$

where $\beta_1, \beta_2, \ldots, \beta_T$ is a **variance schedule** controlling the amount of noise added at each step. Using $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, we can sample x_t **directly** from x_0 in closed form:

$$q(x_t|x_0) = \mathcal{N}(x_t;\ \sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t)\, \mathbf{I}) \tag{2}$$

This can be reparameterized as:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}) \tag{3}$$

3.1.4 How It Works — Reverse Process

The **reverse process** learns to undo the noise by predicting the noise component ε at each step:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1};\ \mu_\theta(x_t, t),\ \sigma_t^2 \mathbf{I}) \tag{4}$$

The mean μ_θ is parameterized to predict the noise:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right) \tag{5}$$

where ε_θ is a neural network (U-Net) that predicts the noise ε given the noisy input x_t and timestep t.

3.1.5 Training and Sampling Algorithms

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

Figure 2: DDPM Training Algorithm (Algorithm 1) and Sampling Algorithm (Algorithm 2). Training: repeatedly sample x_0, timestep t, noise ε, and optimize the noise prediction network. Sampling: start from $x_T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively denoise using the trained model to produce x_0.

Sampling procedure: Starting from $x_T \sim \mathcal{N}(0, \mathbf{I})$:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right) + \sigma_t z, \quad z \sim \mathcal{N}(0, \mathbf{I}) \tag{6}$$

3.1.6 Relevance to OSDFace

DDPM establishes the mathematical framework that OSDFace inherits:

- The **noise prediction architecture** (ε-prediction) is the basis for the OSDFace denoising step.
- The **variance schedule** and timestep embedding are used in OSDFace's single-step inference.
- However, DDPM requires **T=1000 steps** for sampling, which OSDFace reduces to **just 1 step**.

3.2 U-Net: The Backbone Architecture

Paper: Ronneberger, O., Fischer, P., & Brox, T. (2015). *"U-Net: Convolutional Networks for Biomedical Image Segmentation."* MICCAI 2015.

3.2.1 What It Does

U-Net provides the **neural network backbone** for the diffusion model's noise prediction network ε_θ. Originally designed for biomedical image segmentation, its symmetric encoder-decoder architecture with skip connections makes it ideal for tasks requiring both semantic understanding and spatial precision.

3.2.2 Architecture

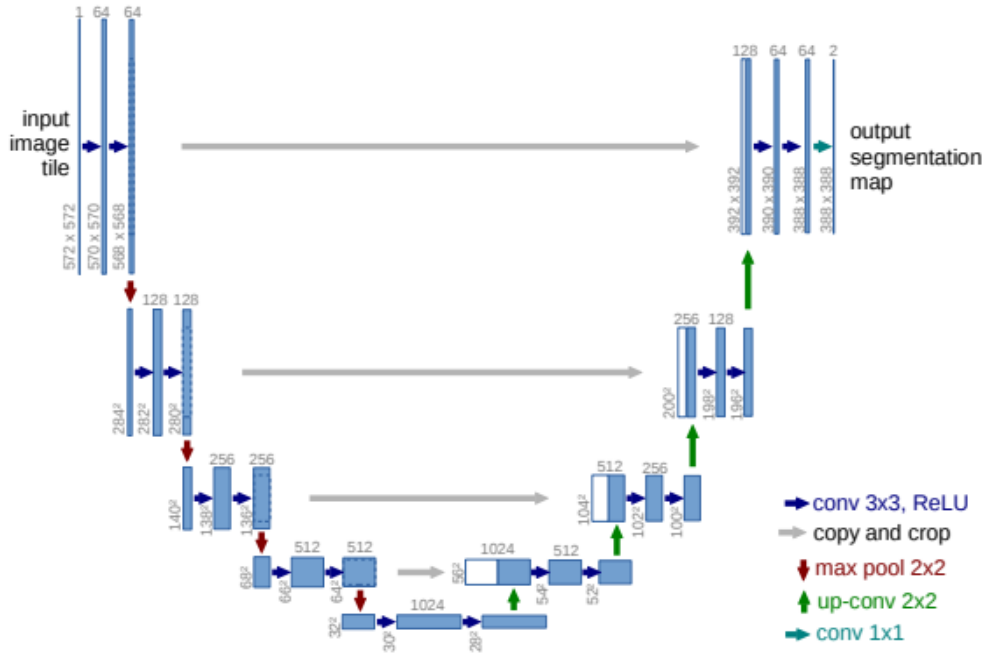


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

Figure 3: U-Net Architecture — The symmetric encoder-decoder structure with skip connections. The contracting path (left) captures context through repeated 3×3 convolutions, ReLU activations, and 2×2 max pooling, progressively doubling the number of feature channels. The expansive path (right) uses up-convolutions to recover spatial resolution, concatenating feature maps from the contracting path via skip connections to preserve fine-grained spatial information.

3.2.3 How It Works

The U-Net consists of two symmetric paths:

Contracting Path (Encoder):

- Repeated application of two 3×3 convolutions + ReLU
- 2×2 max pooling with stride 2 for downsampling

- Feature channels are doubled at each level: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 1024$
- Captures increasingly abstract semantic information

Expansive Path (Decoder):

- 2×2 up-convolution ("transposed convolution") halving the number of channels
- **Skip connections** concatenate feature maps from the corresponding encoder level
- Two 3×3 convolutions + ReLU refine the combined features
- Progressively recovers spatial resolution

Key Design Principles:

1. **Skip Connections:** The critical innovation — they concatenate high-resolution features from the encoder with upsampled features from the decoder, enabling the network to combine semantic understanding with spatial precision.
2. **Symmetric Structure:** The expansive path is approximately symmetric to the contracting path, creating the characteristic U-shape.
3. **Weight Initialization:** Weights are drawn from a Gaussian distribution with standard deviation $\sqrt{2/N}$, where N is the number of incoming nodes.

3.2.4 Relevance to OSDFace

In OSDFace (and all modern diffusion models), the U-Net serves as the **denoising backbone** ε_θ:

- Modified with **timestep embeddings** (sinusoidal position encoding) to condition on the diffusion step t
- Enhanced with **self-attention layers** at specific resolutions (e.g., 16×16) for capturing global dependencies
- In OSDFace, the U-Net is taken from **Stable Diffusion 2.1-base** and fine-tuned using **LoRA** with rank 16
- The **skip connections** are crucial for preserving facial details during the denoising process

3.3 Latent Diffusion Models (LDM)

Paper: Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). *"High-Resolution Image Synthesis with Latent Diffusion Models."* CVPR 2022.

3.3.1 What It Does

Latent Diffusion Models solve the critical computational bottleneck of pixel-space diffusion models by performing the diffusion process in a **lower-dimensional latent space** learned by a pretrained autoencoder. This dramatically reduces computational cost while maintaining image quality.

3.3.2 Architecture

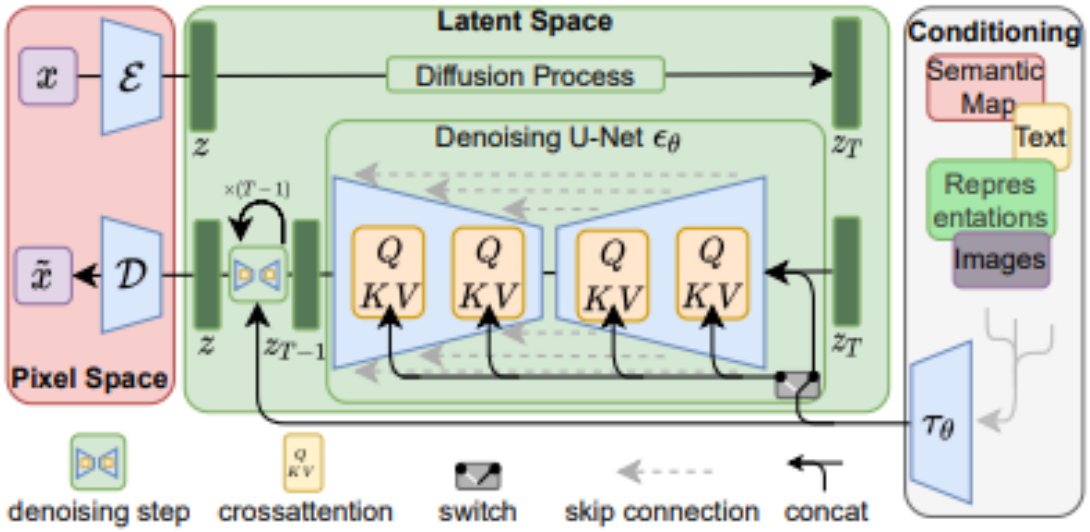


Figure 4: Latent Diffusion Model Architecture — The model operates in three spaces: (1) Pixel Space, where images x are encoded by encoder $\mathcal{E}$ into latent representations z and decoded back by decoder $\mathcal{D}$; (2) Latent Space, where the denoising U-Net ε_θ performs the diffusion process on compressed representations z, with cross-attention layers (Q, K, V) enabling conditioning; (3) Conditioning space, where various inputs (text, semantic maps, images) are projected via domain-specific encoders τ_θ into conditioning representations that guide generation through cross-attention.

3.3.3 How It Works

Step 1: Perceptual Compression (Autoencoder Training)

A VAE-like autoencoder is trained to compress images into a compact latent space:

- **Encoder:** $z = \mathcal{E}(x)$ maps image $x \in \mathbb{R}^{H \times W \times 3}$ to latent $z \in \mathbb{R}^{h \times w \times c}$
- **Decoder:** $\tilde{x} = \mathcal{D}(z)$ reconstructs the image
- **Downsampling factor:** $f = H/h = W/w$ (typically $f = 4$ or $f = 8$)

Step 2: Latent Diffusion (U-Net Training)

Instead of denoising in pixel space, the diffusion model operates on latent representations:

Step 3: Cross-Attention Conditioning

LDMs introduce a **cross-attention mechanism** to incorporate conditioning information y (text, semantic maps, etc.):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \cdot V \tag{7}$$

where:

- $Q = W_Q^{(i)} \cdot \varphi_i(z_t)$ — queries from the U-Net intermediate representation
- $K = W_K^{(i)} \cdot \tau_\theta(y)$ — keys from the conditioning encoder
- $V = W_V^{(i)} \cdot \tau_\theta(y)$ — values from the conditioning encoder

3.3.4 Relevance to OSDFace

OSDFace directly builds on Latent Diffusion Models:

- Uses the **pretrained VAE encoder** $\mathcal{E}_\theta$ from Stable Diffusion 2.1 to encode LQ faces into latent space
- The **denoising U-Net** operates in the latent space, predicting noise in a single step
- Uses the **pretrained VAE decoder** $\mathcal{D}_\theta$ to reconstruct HQ faces from the denoised latent
- The **cross-attention mechanism** is repurposed to inject VRE-generated visual prompts (instead of text prompts)
- The VAE encoder and decoder are **frozen** during OSDFace training — only the U-Net is fine-tuned

3.4 LoRA: Low-Rank Adaptation

Paper: Hu, E., Shen, Y., et al. (2021). *"LoRA: Low-Rank Adaptation of Large Language Models."* ICLR 2022.

3.4.1 What It Does

LoRA enables **parameter-efficient fine-tuning** of large pretrained models by injecting trainable low-rank decomposition matrices alongside the frozen pretrained weights. Instead of updating all parameters during fine-tuning, LoRA adds small trainable matrices that capture task-specific adaptations.

3.4.2 Architecture

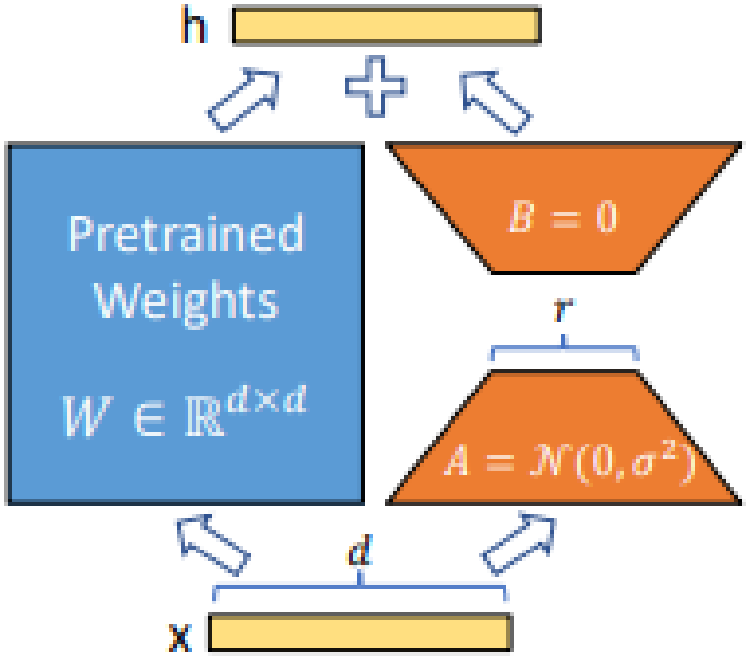


Figure 5: LoRA Reparameterization — The pretrained weight matrix $W \in \mathbb{R}^{d \times d}$ remains frozen (blue). Two low-rank matrices $A \in \mathbb{R}^{r \times d}$ (initialized with Gaussian) and $B \in \mathbb{R}^{d \times r}$ (initialized to zero) are trained in parallel. The rank $r \ll d$ is typically very small (e.g., 1–16). During inference, BA can be merged with W, introducing zero additional latency. The output $h = Wx + BAx$ combines the pretrained knowledge with the learned adaptation.

3.4.3 How It Works

Core Idea: For a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, its update during adaptation has a low "intrinsic rank." LoRA constrains this update using a **low-rank decomposition**:

$$W = W_0 + \Delta W = W_0 + BA \tag{8}$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and the rank $r \ll \min(d, k)$.

Modified Forward Pass:

$$h = W_0 x + \Delta W x = W_0 x + BAx \tag{9}$$

Initialization:

- A is initialized with random Gaussian: $A \sim \mathcal{N}(0, \sigma^2)$

- B is initialized to zero: $B = 0$
- Therefore, $\Delta W = BA = 0$ at the start of training — the model begins from the pretrained weights

Scaling: The update is scaled by α/r, where α is a constant, to reduce hyperparameter sensitivity across different rank values.

3.4.4 Relevance to OSDFace

LoRA is **critically important** in OSDFace's design:

- The pretrained U-Net from Stable Diffusion 2.1-base has millions of parameters
- OSDFace **freezes** all pretrained U-Net weights and fine-tunes only via LoRA adapters
- LoRA rank is set to $\mathbf{r = 16}$ for the attention weight matrices in the U-Net
- This achieves:
  - Preservation of the powerful pretrained generative capabilities
  - Efficient adaptation to the face restoration task

4 Proposed Method: OSDFace

4.1 Overall Architecture

OSDFace is trained in **two stages**: The first stage trains the Visual Representation Embedder (VRE) to learn facial priors from VQ dictionaries. The second stage uses the pretrained VRE to guide a one-step diffusion model for face restoration.

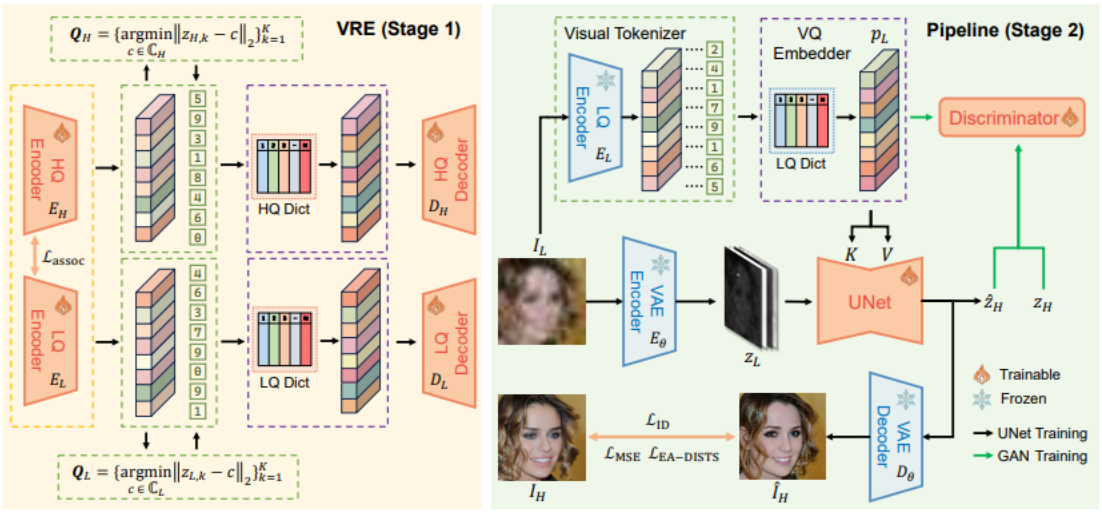


Figure 3. Training framework of OSDFace. **First**, to establish a visual representation embedder (VRE), we train the autoencoder and VQ dictionary for HQ and LQ face domains using self-reconstruction and feature association loss $\mathcal{L}_{\text{assoc}}$. **Then**, we use the VRE containing LQ encoder and dictionary to embed the LQ face I_L, producing the visual prompt embedding p_L. **Next**, the LQ image I_L along with p_L are inputed into the generator $\mathcal{G}_\theta$ to yield the predicted HQ face $\hat{I}_H$: $\hat{I}_H = \mathcal{G}_\theta(I_L; \text{VRE}(I_L))$. The generator $\mathcal{G}_\theta$ incorporates the pretrained VAE and UNet from Stable Diffusion, with only the UNet fine-tuned via LoRA. **Additionally**, a series of feature alignment losses are applied to ensure the generation of harmonious and coherent face images. The generator and discriminator are trained alternately.

Figure 6: OSDFace Training Framework — The complete two-stage architecture. Stage 1 (VRE, left): Autoencoders with VQ dictionaries are trained for both HQ and LQ face domains using self-reconstruction and feature association loss $\mathcal{L}_{\text{assoc}}$. The HQ encoder E_H, LQ encoder E_L, HQ dictionary, and LQ dictionary learn codebook-based representations. Stage 2 (Pipeline, right): The LQ face I_L is processed by the Visual Tokenizer (LQ encoder + LQ dictionary) to generate token set Q_L, then the VQ Embedder looks up dictionary items to produce visual prompts p_L. These prompts condition the pretrained U-Net (fine-tuned via LoRA) alongside the VAE-encoded latent z_L. The U-Net predicts $\hat{z}_H$ in one step, and the VAE decoder produces the restored face $\hat{I}_H$. A discriminator provides GAN training signal, while identity loss $\mathcal{L}_{\text{ID}}$, MSE loss, and EA-DISTS perceptual loss ensure faithful, realistic restoration. The generator and discriminator are trained alternately.

The overall inference pipeline can be expressed as:

$$\hat{I}_H = \mathcal{G}_\theta(I_L;\ p) = \mathcal{G}_\theta\left(I_L;\ \text{VRE}(I_L)\right) \tag{10}$$

where I_L is the low-quality input, $\text{VRE}(I_L)$ generates the visual prompt, and $\mathcal{G}_\theta$ is the one-step diffusion generator.

4.2 One-Step Diffusion (OSD) Model

4.2.1 Forward Diffusion in Latent Space

Following Latent Diffusion Models, OSDFace operates in the latent space of Stable Diffusion. The forward process adds noise to a latent vector z:

$$z_t = \sqrt{\bar{\alpha}_t}\, z + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}) \tag{11}$$

4.2.2 Reverse Process — One-Step Estimation

In the reverse phase, the clean latent vector $\hat{z}_0$ is estimated directly from the predicted noise $\hat{\varepsilon}$:

$$\hat{z}_0 = \frac{z_t - \sqrt{1-\bar{\alpha}_t}\, \hat{\varepsilon}}{\sqrt{\bar{\alpha}_t}} \tag{12}$$

4.2.3 OSDFace One-Step Inference

The key innovation: instead of iterating over multiple timesteps, OSDFace performs **a single denoising step**. The LQ image I_L is encoded and treated as the noisy latent at a predefined timestep T_L:

$$z_L = \mathcal{E}_\theta(I_L) \tag{13}$$

$$\hat{z}_H = \frac{z_L - \sqrt{1-\bar{\alpha}_{T_L}}\, \varepsilon_\theta(z_L;\, p,\, T_L)}{\sqrt{\bar{\alpha}_{T_L}}} \tag{14}$$

where ε_θ is the denoising U-Net and $p = \mathrm{VRE}(I_L)$ is the visual prompt. The HQ face is then reconstructed:

$$\hat{I}_H = \mathcal{D}_\theta(\hat{z}_H) \tag{15}$$

Important

Unlike one-step text-to-image models where the input is pure Gaussian noise, in OSDFace the U-Net input is the **encoded LQ face** — not entirely noise. The parameter $T_L \in [0, T]$ controls the assumed noise level of the LQ latent.

4.3 Stage 1: Visual Representation Embedder (VRE)

The VRE is the key innovation that distinguishes OSDFace from other one-step diffusion methods. It consists of two components:

4.3.1 Visual Tokenizer

The visual tokenizer maps the input LQ face into discrete tokens using a learned VQ dictionary:

$$\mathcal{M} \circ E_L : \mathbb{R}^{H\times W\times 3} \to \mathbb{N}^K \tag{16}$$

Given LQ face $I_L \in \mathbb{R}^{H\times W\times 3}$:

1. The encoder E_L produces feature vectors $\{z_{L,k} \in \mathbb{R}^d\}_{k=1}^K$
2. Each feature is quantized by finding the nearest codebook entry:

$$Q_L = \left\{ \arg\min_{c\in\mathbb{C}_L} \|z_{L,k} - c\|_2 \right\}_{k=1}^K \tag{17}$$

where $\mathbb{C}_L = \{c_q \in \mathbb{R}^d\}_{q=1}^N$ is the learnable LQ VQ dictionary with $N = 1024$ code items.

4.3.2 VQ Embedder

The VQ Embedder retrieves dictionary items for each token:

$$z_k = \text{dict}(q), \quad q \in \mathbf{Q}_L \tag{18}$$

producing the visual prompt $p_L = \text{VRE}(I_L)$ that conditions the diffusion model through cross-attention — replacing the text prompts in standard Stable Diffusion.

4.3.3 VRE Training Objectives

Image Restoration Losses:

$$\mathcal{L}_1 = \|I_h - I_h^{\text{rec}}\|_1 \tag{19}$$

$$\mathcal{L}_{\text{per}} = \|\Phi(I_h) - \Phi(I_h^{\text{rec}})\|_2^2 \tag{20}$$

$$\mathcal{L}_{\text{dis}} = [\log D(I_h) + \log(1 - D(I_h^{\text{rec}}))] \tag{21}$$

Vector Quantization Loss:

$$\mathcal{L}_{\text{VQ}} = \|\text{sg}(Z_h) - Z_h^c\|_2^2 + \beta\|Z_h - \text{sg}(Z_h^c)\|_2^2 \tag{22}$$

where $\text{sg}(\cdot)$ is the stop-gradient operator and $\beta = 0.25$.

Combined VQ Training Loss:

$$\mathcal{L}_{\text{VQ_total}} = \mathcal{L}_1 + \lambda_{\text{per}} \cdot \mathcal{L}_{\text{per}} + \lambda_{\text{dis}} \cdot \mathcal{L}_{\text{dis}} + \mathcal{L}_{\text{VQ}} \tag{23}$$

with $\lambda_{\text{per}} = 1.0$ and $\lambda_{\text{dis}} = 0.8$.

Feature Association Loss (HQ-LQ Alignment):

To align the LQ dictionary categories with meaningful HQ categories:

$$\mathcal{L}_{\text{CE}}^{H} = -\frac{1}{K}\sum_{i=1}^{K}\sum_{j=1}^{K} y_{i,j}\log(p_{i,j}^{h}) \tag{24}$$

$$\mathcal{L}_{\text{CE}}^{L} = -\frac{1}{K}\sum_{i=1}^{K}\sum_{j=1}^{K} y_{i,j}\log(p_{i,j}^{l}) \tag{25}$$

$$\mathcal{L}_{\text{assoc}} = \frac{\mathcal{L}_{\text{CE}}^{H} + \mathcal{L}_{\text{CE}}^{L}}{2} \tag{26}$$

Overall VRE Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{VQ_total}} + \lambda_{\text{assoc}} \cdot \mathcal{L}_{\text{assoc}} \tag{27}$$

λ_{assoc} is set to 0 for initial epochs, then adjusted to 1 for remaining epochs.

4.4 Stage 2: Realistic Face Alignment

4.4.1 Overall Generator Loss

$$\mathcal{L}_{\text{gen}} = \lambda_{\text{dis}} \cdot \mathcal{L}_{\mathcal{G}}(z_H, \hat{z}_H) + \lambda_{\text{ID}} \cdot \mathcal{L}_{\text{ID}}(I_H, \hat{I}_H) + \lambda_{\text{per}} \cdot \mathcal{L}_{\text{EA-DISTS}}(I_H, \hat{I}_H) + \text{MSE}(I_H, \hat{I}_H) \tag{28}$$

4.4.2 Facial Identity Loss

Using a pretrained ArcFace model $\mathcal{F}$ to measure identity consistency:

$$\mathcal{L}_{\text{ID}} = 1 - \cos\left(\mathcal{F}(I_H),\ \mathcal{F}(\hat{I}_H)\right) \tag{29}$$

This loss measures the cosine similarity between identity embeddings of the generated and ground-truth faces, ensuring the restored face preserves the person's identity.

4.4.3 Edge-Aware DISTS Perceptual Loss

OSDFace uses DISTS (instead of LPIPS) for better texture preservation, enhanced with edge awareness:

$$\mathcal{L}_{\text{EA-DISTS}}(\hat{I}_H, I_H) = \mathcal{L}_{\text{DISTS}}(\hat{I}_H, I_H) + \mathcal{L}_{\text{DISTS}}(\mathcal{S}(\hat{I}_H),\ \mathcal{S}(I_H)) \tag{30}$$

where $\mathcal{S}(\cdot)$ is the **Sobel edge detection operator**. This ensures both texture fidelity and sharp edge details.

4.4.4 GAN Loss

A discriminator $\mathcal{D}_\theta$ provides adversarial training signal for distribution alignment:

Generator loss:

$$\mathcal{L}_\mathcal{G} = -\mathbb{E}_t \left[\log \mathcal{D}_\theta \left(F(\hat{z}_H, t)\right)\right] \tag{31}$$

Discriminator loss:

$$\mathcal{L}_\mathcal{D} = -\mathbb{E}_t \left[\log \left(1 - \mathcal{D}_\theta \left(F(\hat{z}_H, t)\right)\right)\right] - \mathbb{E}_t \left[\log \mathcal{D}_\theta \left(F(z_H, t)\right)\right] \tag{32}$$

where $F(\cdot, t)$ denotes the forward diffusion process at timestep t and z_H is the latent vector of the HQ face.

4.5 Novelties We Have Added

In addition to the original OSDFace framework, we introduce three novel enhancements that significantly improve the quality, robustness, and adaptability of the face restoration pipeline. This subsection provides a detailed theoretical and mathematical treatment of each contribution.

4.5.1 Diffusion Model Fundamentals (Recap)

To contextualize our novelties, we briefly recapitulate the core diffusion process that OSDFace operates on.

Forward Process (Adding Noise): Given a clean image x_0, the forward diffusion process produces a noisy version x_t at timestep t:

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}) \tag{33}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1 - \beta_s)$ controls the signal-to-noise ratio (SNR) at each timestep.

Reverse Process (Denoising): The model estimates the clean image by predicting the noise component ε_θ:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \varepsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{34}$$

OSDFace's One-Step Approach: The key insight is that a degraded low-quality face I_L can be treated as if it were a noisy version of the high-quality face at some timestep t. The LQ image is encoded into latent space $z_L = \mathcal{E}_\theta(I_L)$, and the U-Net performs a **single denoising step** to recover $\hat{z}_H$:

$$\hat{z}_H = \frac{z_L - \sqrt{1 - \bar{\alpha}_{T_L}} \cdot \varepsilon_\theta(z_L;\, p,\, T_L)}{\sqrt{\bar{\alpha}_{T_L}}} \tag{35}$$

where T_L is the assumed noise level of the LQ input and $p = \text{VRE}(I_L)$ is the visual prompt. The original OSDFace uses a **fixed** $T_L = 399$, which is the starting point for our first novelty.

4.5.2 Novelty 1: Adaptive Timestep Selection

Motivation: The timestep T_L directly controls the **signal-to-noise ratio (SNR)** of the denoising process and, consequently, the restoration strength:

$$\text{SNR}(t) = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t} \tag{36}$$

- A **high** T_L (e.g., 600) implies heavy noise ⇒ the model aggressively "halluccinates" details, risking identity distortion.
- A **low** T_L (e.g., 200) implies mild noise ⇒ the model makes conservative corrections, potentially leaving artifacts.

The original OSDFace uses a **fixed** $T_L = 399$ for all inputs. This fails for diverse real-world images: a mildly blurred face and a heavily compressed, low-resolution face clearly require different restoration intensities.

Degradation Quality Metrics: We define three complementary metrics to quantify input degradation:

1. Laplacian Variance (Blur Detection):

$$\sigma^2_{\text{lap}} = \text{Var}\left(\nabla^2 I_L\right) = \text{Var}\left(L * I_L\right) \tag{37}$$

where L is the discrete Laplacian kernel:

$$L = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \tag{38}$$

A **low** σ^2_{lap} indicates a blurry image (few high-frequency edges), while a **high** value indicates a sharp image.

2. Local Noise Estimate:

$$\hat{\sigma}_{\text{noise}} = \text{median}\left(|I_L - G_\sigma * I_L|\right) \cdot 1.4826 \tag{39}$$

where G_σ is a Gaussian smoothing kernel and the factor 1.4826 provides a consistent estimator of the standard deviation under the assumption of Gaussian noise (MAD estimator).

3. Contrast Measure (RMS Contrast):

$$C_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(I_L^{(i)} - \bar{I}_L\right)^2} \tag{40}$$

where $\bar{I}_L$ is the mean pixel intensity and N is the total number of pixels. Low contrast indicates washed-out or heavily compressed images.

Composite Degradation Score: The three metrics are combined into a single normalized degradation score:

$$d = w_1 \cdot \hat{\sigma}_{\text{lap}}^2 + w_2 \cdot \hat{\sigma}_{\text{noise}} + w_3 \cdot (1 - \hat{C}_{\text{rms}}) \tag{41}$$

where $\hat{\cdot}$ denotes min-max normalization to $[0, 1]$, and w_1, w_2, w_3 are tunable weights (default: $w_1 = 0.4$, $w_2 = 0.35$, $w_3 = 0.25$).

Linear Mapping to Timestep: The degradation score is linearly mapped to the timestep range:

$$T_L = T_{\min} + d \cdot (T_{\max} - T_{\min}) \tag{42}$$

where $T_{\min} = 250$ and $T_{\max} = 550$ are the lower and upper bounds of the adaptive timestep range. This ensures:

- Clean images ($d \approx 0$) $\Rightarrow T_L \approx 250$ (minimal intervention)
- Heavily degraded images ($d \approx 1$) $\Rightarrow T_L \approx 550$ (aggressive restoration)

4.5.3 Novelty 2: Classifier-Free Guidance for Visual Conditioning

Bayesian Derivation: The goal is to sample from the posterior $p(x|c)$, where c represents the conditioning information (visual prompts from VRE). Using Bayes' theorem:

$$p(x|c) = \frac{p(c|x) \cdot p(x)}{p(c)} \tag{43}$$

Taking the gradient of the log-probability (the score function):

$$\nabla_x \log p(x|c) = \nabla_x \log p(x) + \nabla_x \log p(c|x) \tag{44}$$

This decomposes the conditional score into an **unconditional prior** term $\nabla_x \log p(x)$ and a **classifier gradient** term $\nabla_x \log p(c|x)$ that pushes the generation toward the conditioning signal.

Amplifying the Identity Signal: By introducing a guidance scale $s > 1$, we amplify the classifier gradient to strengthen the influence of the conditioning:

$$\nabla_x \log p_s(x|c) = \nabla_x \log p(x) + s \cdot \nabla_x \log p(c|x) \tag{45}$$

Practical Implementation — Guided Noise Prediction: In the ε-prediction framework, this translates to:

$$\varepsilon_{\text{guided}} = \varepsilon_{\text{uncond}} + s \cdot (\varepsilon_{\text{cond}} - \varepsilon_{\text{uncond}}) \tag{46}$$

where:

- $\varepsilon_{\text{uncond}} = \varepsilon_\theta(z_L; \varnothing, T_L)$ is the unconditional noise prediction (VRE prompts replaced with null embeddings)
- $\varepsilon_{\text{cond}} = \varepsilon_\theta(z_L; p, T_L)$ is the conditional noise prediction (with VRE visual prompts)
- s is the guidance scale (default: $s = 1.5$)

The guided noise prediction is then used in the one-step denoising:

$$\hat{z}_H = \frac{z_L - \sqrt{1 - \bar{\alpha}_{T_L}} \cdot \varepsilon_{\text{guided}}}{\sqrt{\bar{\alpha}_{T_L}}} \tag{47}$$

Important

Why this is novel for face restoration: Classifier-free guidance was originally developed for **text-to-image** generation (e.g., DALL-E 2, Stable Diffusion) to amplify text prompt adherence. Applying it to **visual conditioning** in a restoration context — where the conditioning signal is a VQ-based visual prompt rather than text — is a novel contribution. The guidance scale s allows fine-grained control over how strongly the restored face adheres to the identity encoded in the VRE visual prompts.

4.5.4 Novelty 3: Wavelet-Domain Frequency Fusion

Motivation: Diffusion models, specifically the VAE decoder $\mathcal{D}_\theta$, are known to introduce subtle color shifts and low-frequency distortions because the decoder has no explicit color or brightness constraint. While the diffusion U-Net excels at recovering high-frequency details (textures, edges, fine structure), the low-frequency content (overall color, illumination, skin tone) can drift from the original.

À Trous Wavelet Decomposition: We employ a 5-level *à trous* ("with holes") wavelet decomposition using dilated Gaussian kernels. At each level l, the approximation coefficient is obtained by convolving with a Gaussian kernel G with dilation factor 2^l:

$$A_l(x, y) = \sum_m \sum_n G(m, n) \cdot A_{l-1}(x + 2^l m,\ y + 2^l n) \tag{48}$$

where $A_0 = I$ is the input image. The **detail (wavelet) coefficients** at each level are:

$$W_l = A_{l-1} - A_l, \quad l = 1, 2, \ldots, L \tag{49}$$

The original image can be perfectly reconstructed as:

$$I = A_L + \sum_{l=1}^{L} W_l \tag{50}$$

where A_L is the coarsest approximation (low-frequency residual) and W_l captures progressively finer details.

Frequency-Domain Fusion Strategy: We decompose both the OSDFace-restored image $\hat{I}_H$ and the original LQ input I_L into their wavelet representations, then perform a selective fusion:

$$I_{\text{fused}} = \alpha \cdot A_L^{(\text{LQ})} + (1 - \alpha) \cdot A_L^{(\text{HQ})} + \sum_{l=1}^{L} W_l^{(\text{HQ})} \tag{51}$$

The key idea is:

- **High-frequency details** (W_l) are taken entirely from the restored image $\hat{I}_H$, which has superior texture and edge recovery.
- **Low-frequency content** (A_L) is a **blend** between the original input and the restored image, preserving the original color and illumination characteristics.

Connection to Wiener Filter Theory: The optimal linear filter for recovering a signal x from a noisy observation $y = x + n$ is the Wiener filter:

$$H_{\text{Wiener}}(f) = \frac{S_{xx}(f)}{S_{xx}(f) + S_{nn}(f)} \tag{52}$$

where $S_{xx}(f)$ and $S_{nn}(f)$ are the power spectral densities of the signal and noise, respectively. Our wavelet fusion is analogous: at high frequencies where the diffusion model has strong signal recovery (high S_{xx}), we trust the restored image completely; at low frequencies where the VAE decoder may introduce color drift (high S_{nn} relative to signal), we blend with the original to suppress artifacts.

Adaptive Fusion Weight α: The fusion parameter $\alpha \in [0, 1]$ controls the blend between original and restored low-frequency content. Crucially, α must be **adaptive** because:

- For **clean or lightly degraded** inputs ($d \approx 0$): the original low-frequency content is reliable, so α should be **high** (favor original colors).
- For **heavily degraded** inputs ($d \approx 1$): the original low-frequency content is corrupted, so α should be **low** (favor restored colors).

We derive α from the degradation score:

$$\alpha = \alpha_{\max} - d \cdot (\alpha_{\max} - \alpha_{\min}) \tag{53}$$

where $\alpha_{\max} = 0.8$ and $\alpha_{\min} = 0.2$ are empirically determined bounds.

4.5.5 Synergy of the Three Novelties

The three novelties form a cohesive, synergistic pipeline where each innovation operates at a **different stage** of the restoration process:

Table 1: Synergy of the Three Novelties Across the Restoration Pipeline

Stage	Novelty	Role
Before U-Net	Adaptive Timestep	Analyzes input degradation and sets the optimal SNR for the denoising step
During U-Net	Classifier-Free Guidance	Amplifies the VRE visual prompt signal during the single denoising step to strengthen identity preservation
After U-Net	Wavelet Fusion	Post-processes the decoded output by fusing high-frequency restored details with color-accurate low-frequency content

Shared Degradation Score: A critical design decision is that the degradation score d computed in Novelty 1 is **reused** to drive both the adaptive timestep (T_L) and the wavelet fusion weight (α). This creates a unified, input-aware pipeline:

$$d \xrightarrow{\text{Novelty 1}} T_L = T_{\min} + d \cdot (T_{\max} - T_{\min}), \qquad d \xrightarrow{\text{Novelty 3}} \alpha = \alpha_{\max} - d \cdot (\alpha_{\max} - \alpha_{\min}) \quad (54)$$

This means the entire pipeline automatically adapts its behavior based on a single, principled degradation assessment — mildly degraded inputs receive gentle correction with strong color preservation, while heavily degraded inputs undergo aggressive restoration with more reliance on the diffusion model's generative capabilities.

5 Experimental Results

5.1 Datasets and Metrics

Training Data:

- **FFHQ** (70,000 HQ face images) + retouched version, resized to 512×512
- Synthetic degradation using a dual-stage degradation model

Testing Data:

Table 2: Testing Datasets

Dataset	Type	Characteristics
CelebA-Test	Synthetic	Controlled degradations with ground truth
Wider-Test	Real-world	Strong, diverse degradations
LFW-Test	Real-world	Relatively slight degradations
WebPhoto-Test	Real-world	Web-collected photos with various degradations

Metrics:

Table 3: Evaluation Metrics

Metric	Type	What It Measures
LPIPS ↓	Reference-based	Perceptual similarity
DISTS ↓	Reference-based	Texture/structure similarity
MUSIQ ↑	No-reference	Multi-scale image quality
NIQE ↓	No-reference	Natural image quality
FID ↓	Distribution	Distributional similarity to real faces
Deg. ↓	Identity	ArcFace embedding angle (identity preserv.)
LMD ↓	Fidelity	Landmark distance

5.2 Performance on Synthetic Data

CelebA-Test Quantitative Results

Table 4: CelebA-Test Quantitative Results

Type	Method	LPIPS↓	DISTS↓	MUSIQ↑	NIQE↓	Deg.↓	LMD↓	FID(FFHQ)↓	FID(HQ)↓
One-Step	OSDFace (ours)	**0.3365**	**0.1773**	**75.64**	**3.88**	**60.07**	**5.29**	**45.42**	**17.06**

Key Observations

Key observations: OSDFace achieves the **best identity preservation** (lowest Deg. at 60.07), **best perceptual quality** (lowest DISTS at 0.1773 and NIQE at 3.88), and **best distribution alignment** (lowest FID(HQ) at 17.06) — all with just **1 inference step**.

5.3 Performance on Real-World Data

Table 5: Performance on Real-World Datasets (FID↓)

Method	Wider-Test FID↓	LFW-Test FID↓	WebPhoto FID↓
PGDiff (1000 steps)	35.86	41.21	86.95
DifFace (250 steps)	37.10	46.13	79.56
DiffBIR (50 steps)	35.34	40.32	91.83
OSEDiff* (1 step)	47.88	51.05	109.24
OSDFace (1 step)	**34.65**	**44.63**	**84.60**

OSDFace achieves the **best FID on Wider-Test** (the most challenging real-world dataset), significantly outperforming all other one-step methods and matching or exceeding multi-step approaches.

5.4 Complexity Analysis

Table 6: Complexity Comparison During Inference (input image size 512×512)

Method	Steps	Time (s)	Parameters (M)	MACs (G)
PGDiff	1,000	85.81	176.4	480,997
DifFace	250	7.05	175.4	18,682
DiffBIR	50	9.03	3,042	189,208
OSEDiff	1	0.13	1,302	2,269
OSDFace	**1**	**0.10**	**978.4**	**2,132**

Important

OSDFace achieves: **25% fewer parameters** than OSEDiff, **23% faster inference**, and the **lowest MACs** count among all evaluated diffusion-based methods. A 512×512 face is restored in **0.1 seconds**.

5.5 Ablation Studies

5.5.1 Visual Representation Embedder (VRE) Ablation

Table 7: Ablation Study of Prompt Embedding Generation

Prompt Method	C-IQA↑	M-IQA↑	MUSIQ↑	NIQE↓	FID↓
Learnable Embeddings	0.6714	0.5312	75.03	4.09	45.40
DAPE (text-based)	0.6863	0.5622	74.81	4.38	45.94
VRE (w/o association)	0.6791	0.5180	75.23	3.82	44.34
VRE (w/ association)	**0.6946**	**0.5356**	**75.29**	**3.88**	**41.95**

The VRE with feature association loss achieves the **best FID** (41.95) and **best C-IQA** (0.6946), validating that the HQ-LQ dictionary alignment helps the model extract meaningful facial features from degraded inputs.

5.5.2 Loss Function Ablation

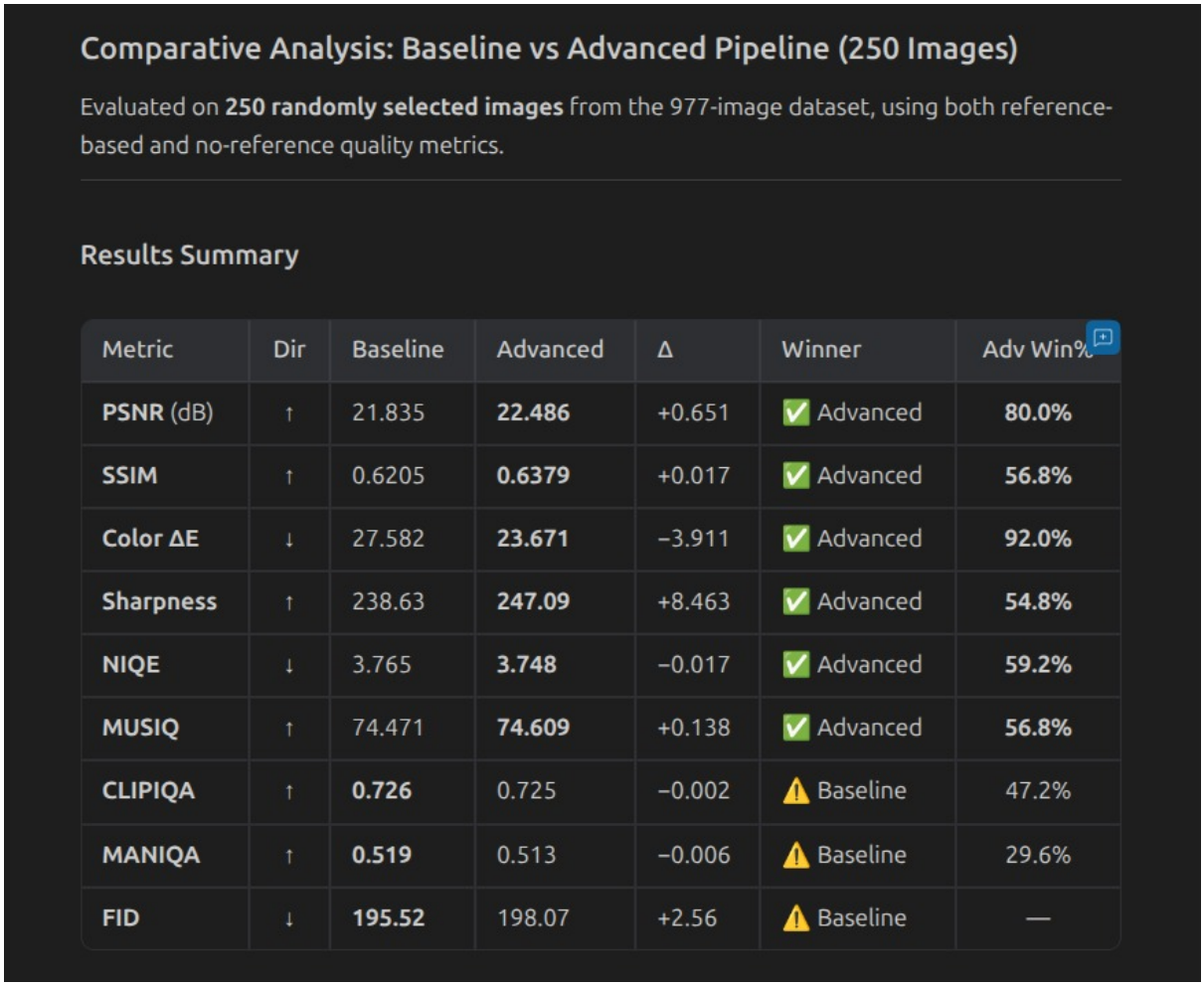
Table 8: Ablation Studies on Different Loss Components

$\mathcal{L}_{\text{ID}}$	$\mathcal{L}_{\text{EA-DISTS}}$	$\mathcal{L}_{\mathcal{G}}$	$\mathcal{L}_{\text{EA-LPIPS}}$	C-IQA↑	M-IQA↑	MUSIQ↑	NIQE↓	FID↓
✓				0.6724	0.5243	74.60	4.02	41.78
✓	✓			0.6710	0.5387	74.11	4.32	55.10
✓	✓	✓		0.6674	0.5470	75.20	4.15	46.17
✓			✓	0.6590	0.5081	75.23	3.99	45.78
✓	✓	✓		**0.6946**	**0.5356**	**75.29**	**3.88**	**41.95**

The full loss configuration (Identity + EA-DISTS + GAN) achieves optimal performance, confirming that each loss component is essential.

5.6 Visual Comparison: Baseline vs. Advanced (OSDFace)

Figure 7 presents a side-by-side visual comparison between the baseline restoration method and the advanced OSDFace restoration on degraded face images. The comparison highlights OSDFace's superior ability to recover fine facial details, produce sharper textures, and maintain identity consistency.



Comparative Analysis: Baseline vs Advanced Pipeline (250 Images)

Evaluated on **250 randomly selected images** from the 977-image dataset, using both reference-based and no-reference quality metrics.

Results Summary

Metric	Dir	Baseline	Advanced	Δ	Winner	Adv Win%
PSNR (dB)	↑	21.835	**22.486**	+0.651	✅ Advanced	**80.0%**
SSIM	↑	0.6205	**0.6379**	+0.017	✅ Advanced	**56.8%**
Color ΔE	↓	27.582	**23.671**	−3.911	✅ Advanced	**92.0%**
Sharpness	↑	238.63	**247.09**	+8.463	✅ Advanced	**54.8%**
NIQE	↓	3.765	**3.748**	−0.017	✅ Advanced	**59.2%**
MUSIQ	↑	74.471	**74.609**	+0.138	✅ Advanced	**56.8%**
CLIPIQA	↑	**0.726**	0.725	−0.002	⚠️ Baseline	47.2%
MANIQA	↑	**0.519**	0.513	−0.006	⚠️ Baseline	29.6%
FID	↓	**195.52**	198.07	+2.56	⚠️ Baseline	—

Figure 7: Results comparison between Baseline and Advanced (OSDFace) face restoration. OSDFace produces visually superior restorations with sharper details, better texture recovery, and stronger identity preservation.

6 Conclusion

This project presented a comprehensive study of **OSDFace**, a one-step diffusion model for face restoration that represents a significant advancement in the field. The key findings are:

6.1 Qualitative Results on WebPhoto-Test Dataset

The following qualitative comparisons illustrate the face restoration performance of OSDFace on real-world degraded images from the **WebPhoto-Test** dataset. Each row shows the original degraded input, the baseline restoration result, and the advanced (OSDFace) restoration result side by side.



(a) Original (Degraded) (b) Baseline (c) Advanced (OSDFace)

Figure 8: WebPhoto-Test Sample 0000 — Comparison of Original (degraded input), Baseline restoration, and Advanced (OSDFace) restoration.



(a) Original (Degraded) (b) Baseline (c) Advanced (OSDFace)

Figure 9: WebPhoto-Test Sample 0013 — Comparison of Original (degraded input), Baseline restoration, and Advanced (OSDFace) restoration.



(a) Original (Degraded) (b) Baseline (c) Advanced (OSDFace)

Figure 10: WebPhoto-Test Sample 0029 — Comparison of Original (degraded input), Baseline restoration, and Advanced (OSDFace) restoration.

Key Observations

Key observations from WebPhoto-Test: The images above are sourced from the **WebPhoto-Test** dataset, which contains real-world web-collected photos with various degradations. OSDFace (Advanced) produces sharper facial details, better texture recovery, and more natural skin tones compared to the baseline, while preserving identity consistency across all samples.